Department of Education
SCHOOLS DIVISION OF CITY OF MEYCAUAYAN
Pag-asa St., Malhacan, City of Meycauayan, Bulacan

Senior High School

12

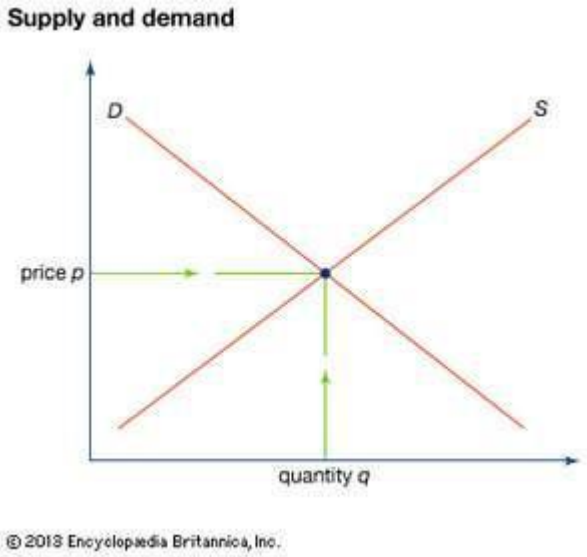
Activity Sheet in

Applied Economics

Quarter 1 – Analyze market demand, market suppy and market equilibrium (ABM_AE12-Ie-h-4)

BASIC ECONOMIC PROBLEMS AND THE PHILIPPINE SOCIOECONOMIC DEVELOPMENT IN THE 21ST CENTURY

LET US KNOW



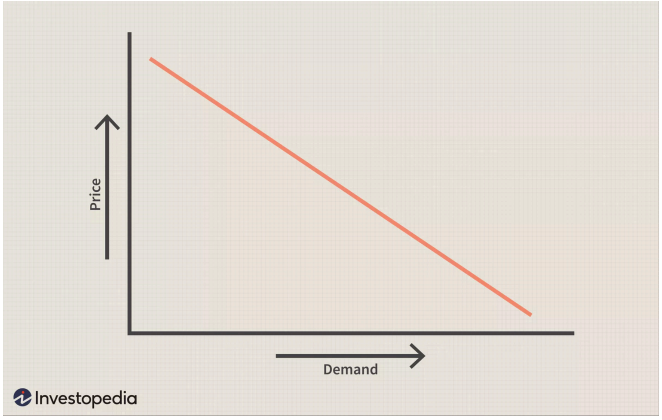
Look at the illustration, what does it show? What do you think are the factors affecting the quantity of a certain product?

This chapter introduces the economic model of demand and supply—one of the most powerful models in all of economics. The discussion here begins by examining how demand and supply determine the price and the quantity sold in markets for goods and services, and how changes in demand and supply lead to changes in prices and quantities. Likewise, the state of equilibrium and how it will be attained, will be discussed.

LET US STUDY

Explaining Demand

Economists use the term **demand** to refer to the amount of some good or service consumers are willing and able to purchase at each price. Demand is based on needs and wants—a consumer may be able to differentiate between a need and a want, but from an economist's perspective they are the same thing. Demand is also based on ability to pay. If you cannot pay for it, you have no effective demand



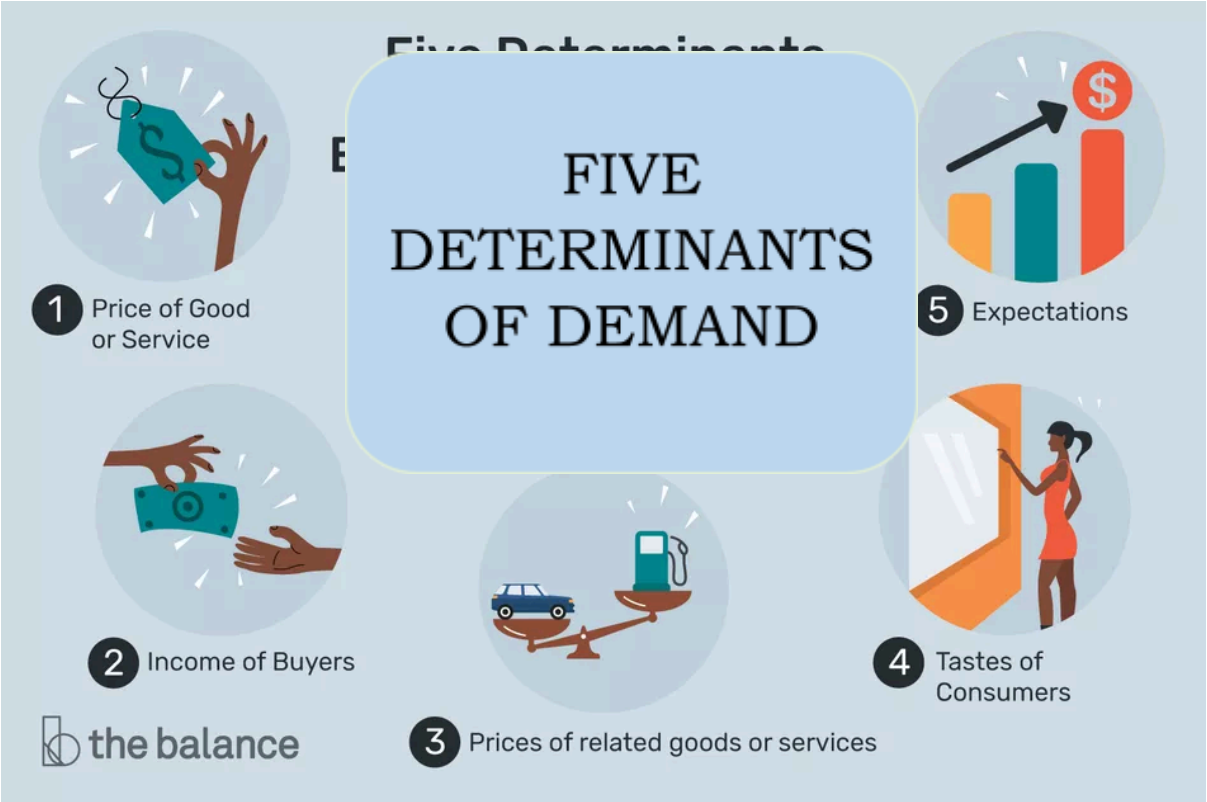
The figure above depicts the most basic relationship between the price of a good and its demand from the standpoint of the consumer. This is actually one of the most important differences between the supply curve and the demand curve. Whereas supply graphs are drawn from the perspective of the producer, demand is portrayed from the perspective of the consumer.

As the price of a good increase, the demand for the product will—except for a few obscure situations—decrease.

WHAT FACTORS AFFECT DEMAND?

These factors matter both for demand by an individual and demand by the market as a whole. Exactly how do these various factors affect demand, and how do we show the effects graphically? To answer those questions, we need the *ceteris paribus* assumption.

A **demand curve** or a **supply curve** is a relationship between two, and only two, variables: quantity on the horizontal axis and price on the vertical axis. The assumption behind a demand curve or a supply curve is that no relevant economic factors, other than the product's price, are changing. Economists call this assumption **ceteris paribus**, a Latin phrase meaning "other things being equal." Any given demand or supply curve is based on the *ceteris paribus* assumption that all else is held equal.



The Five Determinants of Demand

Price

The law of demand states that when prices rise, the quantity of demand falls. That also means that when prices drop, demand will grow. People base their purchasing decisions on price if all other things are equal. The exact quantity bought for each price level is described in the demand schedule. It's then plotted on a graph to show the demand curve.

Income

When income rises, so will the quantity demanded. When income falls, so will demand. But if your income doubles, you won't always buy twice as much of a good or service.

Prices of related goods or services

The price of complementary goods or services raises the cost of using the product you demand, so you'll want less
The opposite reaction occurs when the price of a substitute rises. When that happens, people will want more of the good or service and less of its substitute.

Tastes

When the public's desires, emotions, or preferences change in favor of a product, so does the quantity demand. Likewise, when tastes go against it, that depresses the amount demanded. Brand advertising tries to increase the desire for consumer goods.

Expectations

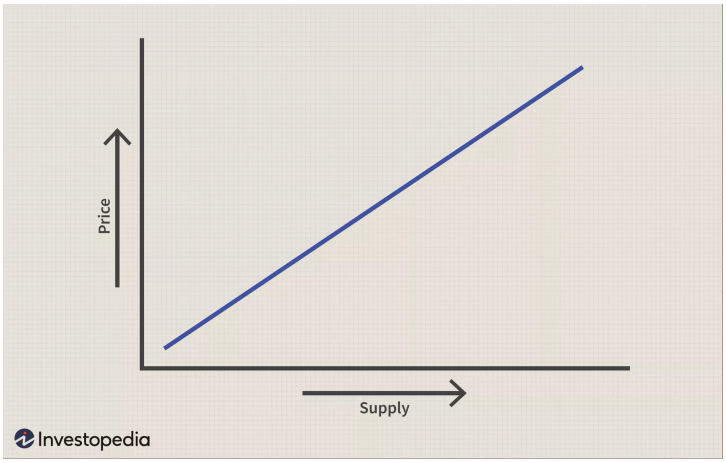
When people expect that the value of something will rise, they demand more of it.

Number of buyers in the market

The number of consumers affects overall, or "aggregate," demand. As more buyers enter the market, demand rises.

Explaining Supply

The supply curve functions in a similar fashion, but it considers the relationship between the price and available supply of an item from the perspective of the producer rather than the consumer.



When prices of a product increase, producers are willing to manufacture more of the product to realize greater profits. Likewise, falling prices depress production as producers may not be able to cover their input costs upon selling the final good.

On the other hand, when prices are higher, producers are encouraged to increase their levels of activity to reap more benefit.

Some of the important factors affecting the supply of a commodity are as follows:

There are several important factors that determine supply of a commodity. A change in any one of these factors will result in a change in supply of the commodity.

1. ***Price of the given Commodity:***

The most important factor determining the supply of a commodity is its price. As a rule, price of a commodity and its supply are directly related. It means, as price increases, the quantity supplied of the given commodity also rises and vice-versa. It happens because at higher prices, there are greater chances of making profit. It induces the firm to offer more for sale in the market.

Supply (S) is a function of price (P) and can be expressed as: S = f (P). The direct relationship between price and supply, known as 'Law of Supply'. The following determinants are termed as 'other factors' or factors other than price'.

2. ***Prices of Other Goods:***

As resources have alternative uses, the quantity supplied of a commodity depends not only on its price, but also on the prices of other commodities. Increase in the prices of other goods makes them more profitable in comparison to the given commodity. As a result, the firm shifts its limited resources from production of the given commodity to production of other goods. For example, increase in the price of other good (say, wheat) will induce the farmer to use land for cultivation of wheat in place of the given commodity (say, rice).

Prices of Factors of Production (inputs):

When the amount payable to factors of production and cost of inputs increases, the cost of production also increases. This decreases the profitability. As a result, seller reduces the supply of the commodity. On the other hand, decrease in prices of factors of production or inputs, increases the supply due to fall in cost of production and subsequent rise in profit margin.
To make ice-cream, firms need various inputs like cream, sugar, machine, labor, etc. When price of one or more of these inputs rises, producing ice-creams will become less profitable and firms supply fewer ice-creams.

3. ***State of Technology:***

Technological changes influence the supply of a commodity. Advanced and improved technology reduces the cost of production, which raises the profit margin. It induces the seller to increase the supply. However, technological degradation or complex and outdated technology will increase the cost of production and it will lead to decrease in supply.

Government Policy (Taxation Policy):

Increase in taxes raises the cost of production and, thus, reduces the supply, due to lower profit margin. On the other hand, tax concessions and subsidies increase the supply as they make it more profitable for the firms to supply goods.

Goals / Objectives of the firm:

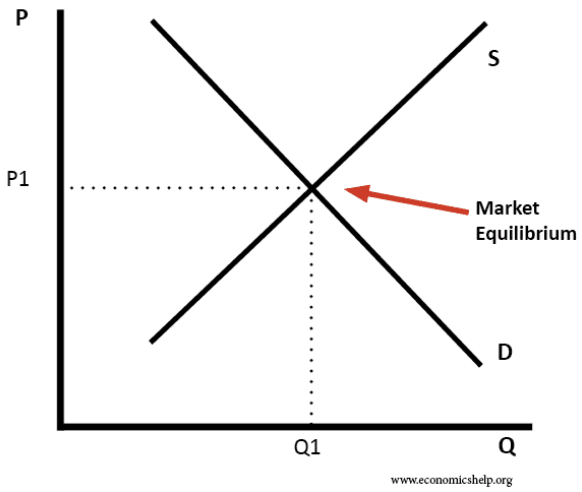
Generally, supply of a commodity increases only at higher prices as it fulfills the objective of profit maximization. However, with change in trend, some firms are willing to supply more even at those prices, which do not maximize their profits. The objective of such firms is to capture extensive markets and to enhance their status and prestige.

Market equilibrium

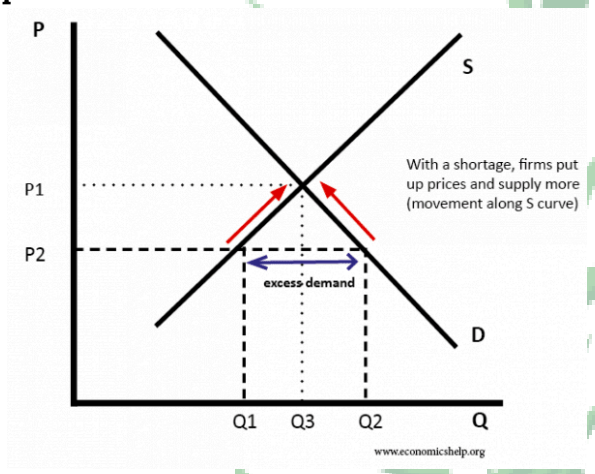
- **Definition of market equilibrium** – A situation where for a particular good supply = demand. When the market is in equilibrium, there is no tendency for prices to change. We say the market-clearing price has been achieved.
- A market occurs where buyers and sellers meet to exchange money for goods.
- The price mechanism refers to how supply and demand interact to set the market price and amount of goods sold.
- At most prices, planned demand does not equal planned supply. This is a state of disequilibrium because there is either a shortage or surplus and firms have an incentive to change the price.

Market equilibrium

Market equilibrium can be shown using supply and demand diagrams
In the diagram below, the equilibrium price is P1. The equilibrium quantity is Q1.

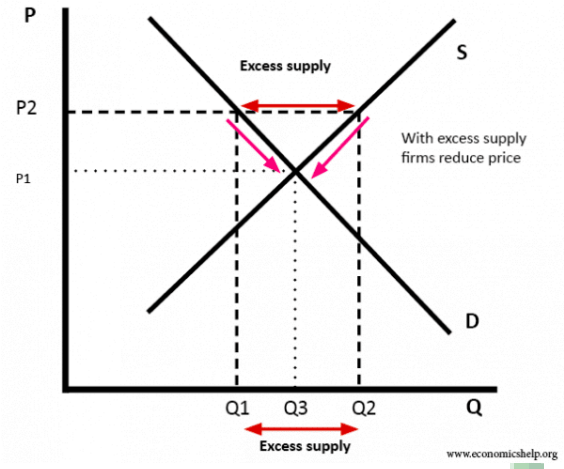


If price is below the equilibrium



- In the above diagram, price (P2) is below the equilibrium. At this price, demand would be greater than the supply. Therefore, there is a shortage of (Q2 – Q1)
- If there is a shortage, firms will put up prices and supply more. As price rises, there will be a movement along the demand curve and less will be demanded.
- Therefore, the price will rise to P1 until there is no shortage and supply = demand.

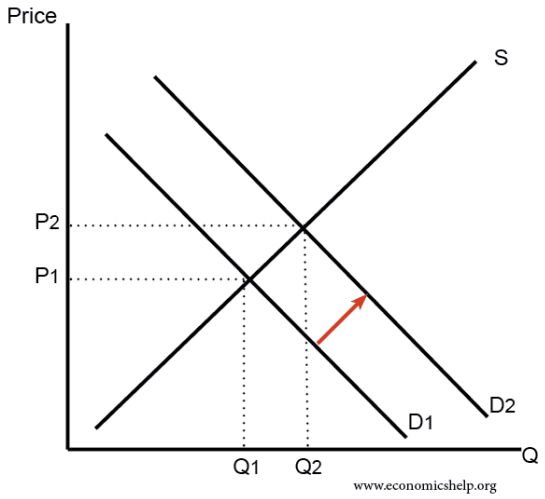
If price is above the equilibrium



- If price was at P2, this is above the equilibrium of P1. At the price of P2, then supply (Q2) would be greater than demand (Q1) and therefore there is too much supply. There is a surplus. (Q2-Q1)
- Therefore, firms would reduce price and supply less. This would encourage more demand and therefore the surplus will be eliminated. The new market equilibrium will be at Q3 and P1.

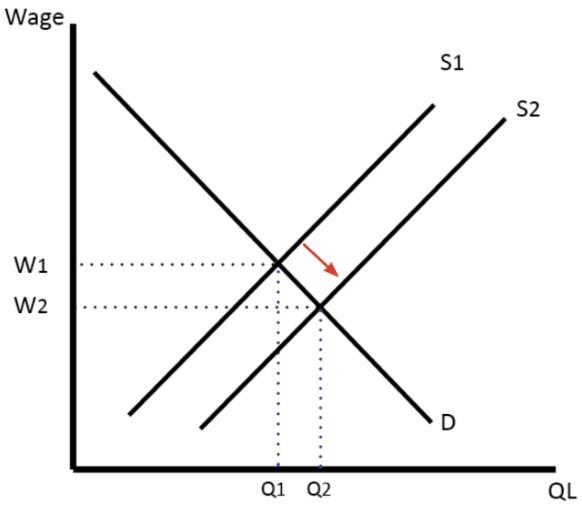
Movements to a new equilibrium

1. Increase in demand



If there was an increase in income the demand curve would shift to the right (D1 to D2). Initially, there would be a shortage of the good. Therefore, the price and quantity supplied will increase leading to a new equilibrium at Q2, P2.

2. Increase in supply



An increase in supply would lead to a lower price and more quantity sold.

LET US PRACTICE (Written Works)

Analyze each situation. Write your answer on space provided

There are 200 buyers (Demand) and 40 jackets (Supply). What will happen to its price? Explain your answer.

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There are 30 buyers (Demand) and 80 jackets (Supply). What will be its effect to the price of the jacket? Explain your answer.

_____

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LET US REMEMBER

KEY TAKEAWAYS

- The law of demand says that at higher prices, buyers will demand less of an economic good.
- The law of supply says that at higher prices, sellers will supply more of an economic good.
- These two laws interact to determine the actual market prices and volume of goods that are traded on a market.
- Several independent factors can affect the shape of market supply and demand, influencing both the prices and quantities that we observe in markets.

LET US APPRECIATE (Performance Task)

1. What is the relationship between demand and supply?

2. How do demand and supply affect the price of commodities? How do this change in price of the basic commodities affect man's purchasing power?

3. What possible remedies or solutions can you suggest limiting the ill effects of the sudden change of the commodity prices?

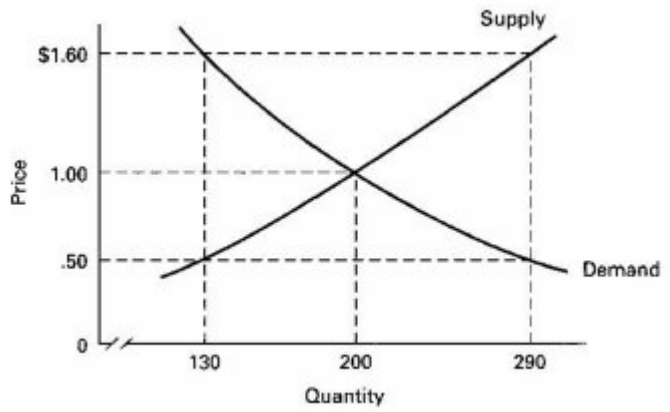
EVALUATION (Performance Task)

Multiple Choice.
Direction. Read each question carefully and choose the letter of the BEST answer from the given choices. Write your answer on the space provided.

_____1. An increase in the price of a product will reduce the amount of it purchased because:

A. supply curves are upsloping
B. the higher price means that real incomes have risen
C. consumers will substitute other products for the one whose price has risen
D. consumers substitute relatively high-priced for relatively low-priced products.

_____2. If the price of N declines, the demand curve for the complementary product Y will:
A. shift to the left
B. decrease
C. shift to the right.
D. remain unchanged.



_____3. Refer to the above diagram. The equilibrium price and quantity in this market will be:

A. 1.00 and 200
B. 1.60 and 130
C. 50 and 130
D. 1.60 and 290

_____4. An increase in the demand for a good will cause

A. an increase in equilibrium price and quantity.
B. a decrease in equilibrium price and quantity.
C. an increase in equilibrium price and a decrease in equilibrium quantity.
D. a decrease in equilibrium price and an increase in equilibrium quantity.

_____5. Market equilibrium refers to a situation in which market price

A. is high enough to allow firms to earn a fair profit.
B. is low enough for consumers to buy all that they want.
C. is at a level where there is neither a shortage nor a surplus.
D. is just above the intersection of the market supply and demand curves.

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